

```
In [53]: import pandas as pd
df = pd.read_csv("C:/Users/mahim/Downloads/Messy_Employee_dataset.csv")
print(df.isnull())
print(df.head())
print(df.shape)
print(df.info())
print(df.columns.tolist())
```

	Employee_ID	First_Name	Last_Name	Age	Department_Region	Status
\						
0	False	False	False	False	False	False
1	False	False	False	True	False	False
2	False	False	False	True	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
...	...	...	...	...	...	...
1015	False	False	False	False	False	False
1016	False	False	False	False	False	False
1017	False	False	False	False	False	False
1018	False	False	False	False	False	False
1019	False	False	False	False	False	False

	Join_Date	Salary	Email	Phone	Performance_Score	Remote_Work
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
...	...	...	...	...	...	...
1015	False	True	False	False	False	False
1016	False	False	False	False	False	False
1017	False	False	False	False	False	False
1018	False	False	False	False	False	False
1019	False	False	False	False	False	False

[1020 rows x 12 columns]

	Employee_ID	First_Name	Last_Name	Age	Department_Region	Status	\
0	EMP1000	Bob	Davis	25.0	DevOps-California	Active	
1	EMP1001	Bob	Brown	NaN	Finance-Texas	Active	
2	EMP1002	Alice	Jones	NaN	Admin-Nevada	Pending	
3	EMP1003	Eva	Davis	25.0	Admin-Nevada	Inactive	
4	EMP1004	Frank	Williams	25.0	Cloud Tech-Florida	Active	

	Join_Date	Salary	Email	Phone	\
0	4/2/2021	59767.65	bob.davis@example.com	-1651623197	
1	7/10/2020	65304.66	bob.brown@example.com	-1898471390	
2	12/7/2023	88145.90	alice.jones@example.com	-5596363211	
3	11/27/2021	69450.99	eva.davis@example.com	-3476490784	
4	1/5/2022	109324.61	frank.williams@example.com	-1586734256	

	Performance_Score	Remote_Work
0	Average	True
1	Excellent	True
2	Good	True
3	Good	True
4	Poor	False

(1020, 12)

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 1020 entries, 0 to 1019

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	Employee_ID	1020 non-null	object
1	First_Name	1020 non-null	object
2	Last_Name	1020 non-null	object
3	Age	809 non-null	float64
4	Department_Region	1020 non-null	object
5	Status	1020 non-null	object

```

6   Join_Date      1020 non-null   object
7   Salary         996 non-null   float64
8   Email          1020 non-null   object
9   Phone          1020 non-null   int64
10  Performance_Score 1020 non-null   object
11  Remote_Work     1020 non-null   bool

```

dtypes: bool(1), float64(2), int64(1), object(8)

memory usage: 88.8+ KB

None

```
['Employee_ID', 'First_Name', 'Last_Name', 'Age', 'Department_Region', 'Status', 'Join_Date', 'Salary', 'Email', 'Phone', 'Performance_Score', 'Remote_Work']
```

```

In [54]: df.rename(columns={
            "Employee_ID": "employee_id",
            "First_Name": "first_name",
            "Last_Name": "last_name",
            "Age": "age",
            "Department_Region": "department_region",
            "Status": "status",
            "Join_Date": "join_date",
            "Salary": "salary",
            "Email": "email",
            "Phone": "phone",
            "Performance_Score": "performance_score",
            "Remote_Work": "remote_work"
        }, inplace=True)
print(df.columns.tolist())

```

```
['employee_id', 'first_name', 'last_name', 'age', 'department_region', 'status', 'join_date', 'salary', 'email', 'phone', 'performance_score', 'remote_work']
```

In []:

```

In [55]: df["age"] = df["age"].fillna(df["age"].median())
df["salary"] = df["salary"].fillna(df["salary"].mean())
print(df["age"].head())
print(df["salary"].head())

```

```
0    25.0
```

```
1    30.0
```

```
2    30.0
```

```
3    25.0
```

```
4    25.0
```

Name: age, dtype: float64

```
0    59767.65
```

```
1    65304.66
```

```
2    88145.90
```

```
3    69450.99
```

```
4   109324.61
```

Name: salary, dtype: float64

```
In [56]: print(df.isnull())
```

	employee_id	first_name	last_name	age	department_region	status
\						
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
...	...	...	...	...	...	...
1015	False	False	False	False	False	False
1016	False	False	False	False	False	False
1017	False	False	False	False	False	False
1018	False	False	False	False	False	False
1019	False	False	False	False	False	False

	join_date	salary	email	phone	performance_score	remote_work
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
...	...	...	...	...	...	...
1015	False	False	False	False	False	False
1016	False	False	False	False	False	False
1017	False	False	False	False	False	False
1018	False	False	False	False	False	False
1019	False	False	False	False	False	False

[1020 rows x 12 columns]

Create Location Table

```
In [57]: df[["department", "location"]] = (
    df["department_region"]
    .str.split("-", n=1, expand=True)
)

df["department"] = df["department"].str.strip()
df["location"] = df["location"].str.strip()
location_table = (
    df[["department", "location"]]
    .drop_duplicates()
    .reset_index(drop=True)
)

print("\nLocation Table:")
print(location_table.head())

# -----
#
# -----
```

Location Table:

	department	location
0	DevOps	California
1	Finance	Texas
2	Admin	Nevada
3	Cloud Tech	Florida
4	Sales	Texas

Merge Location Table

```
In [58]: merged_df = pd.merge(
    df,
    location_table,
    on="department",
    how="left"
)
print(merged_df)
```

	employee_id	first_name	last_name	age	department_region	status	\
0	EMP1000	Bob	Davis	25.0	DevOps-California	Active	
1	EMP1000	Bob	Davis	25.0	DevOps-California	Active	
2	EMP1000	Bob	Davis	25.0	DevOps-California	Active	
3	EMP1000	Bob	Davis	25.0	DevOps-California	Active	
4	EMP1000	Bob	Davis	25.0	DevOps-California	Active	
...	...	...	...	...	...	...	
6115	EMP2019	Heidi	Jones	30.0	DevOps-Illinois	Pending	
6116	EMP2019	Heidi	Jones	30.0	DevOps-Illinois	Pending	
6117	EMP2019	Heidi	Jones	30.0	DevOps-Illinois	Pending	
6118	EMP2019	Heidi	Jones	30.0	DevOps-Illinois	Pending	
6119	EMP2019	Heidi	Jones	30.0	DevOps-Illinois	Pending	

	join_date	salary	email	phone	\
0	4/2/2021	59767.65	bob.davis@example.com	-1651623197	
1	4/2/2021	59767.65	bob.davis@example.com	-1651623197	
2	4/2/2021	59767.65	bob.davis@example.com	-1651623197	
3	4/2/2021	59767.65	bob.davis@example.com	-1651623197	
4	4/2/2021	59767.65	bob.davis@example.com	-1651623197	
...	...	...	...	...	
6115	2/22/2021	77764.24	heidi.jones@example.com	-7629745492	
6116	2/22/2021	77764.24	heidi.jones@example.com	-7629745492	
6117	2/22/2021	77764.24	heidi.jones@example.com	-7629745492	
6118	2/22/2021	77764.24	heidi.jones@example.com	-7629745492	
6119	2/22/2021	77764.24	heidi.jones@example.com	-7629745492	

	performance_score	remote_work	department	location_x	location_y
0	Average	True	DevOps	California	California
1	Average	True	DevOps	California	New York
2	Average	True	DevOps	California	Illinois
3	Average	True	DevOps	California	Florida
4	Average	True	DevOps	California	Texas
...	...	...	...	...	...
6115	Average	False	DevOps	Illinois	New York
6116	Average	False	DevOps	Illinois	Illinois
6117	Average	False	DevOps	Illinois	Florida
6118	Average	False	DevOps	Illinois	Texas
6119	Average	False	DevOps	Illinois	Nevada

[6120 rows x 15 columns]

Average Salary By Department

```
In [59]: avg_salary = (
    merged_df.groupby("department")["salary"]
    .mean()
    .reset_index()
)
```

```

avg_salary.rename(
    columns={"salary": "average_salary"},
    inplace=True
)

print("\nAverage Salary By Department:")
print(avg_salary)

```

```

Average Salary By Department:
  department  average_salary
0      Admin    85190.812361
1  Cloud Tech    84930.982137
2     DevOps    85993.246730
3     Finance    82274.898016
4         HR    85466.190851
5      Sales    86867.327804

```

```

In [60]: import pandas as pd
import matplotlib.pyplot as plt
df = pd.DataFrame({'salary': [35000, 42000, 45000, 48000, 50000, 52000, 55000, 60000, 65000, 90000]})
print(df)
plt.figure(figsize=(6, 4))
plt.boxplot(df['salary'])
plt.title('Box Plot of Salary')
plt.ylabel('Salary')
plt.show()

```

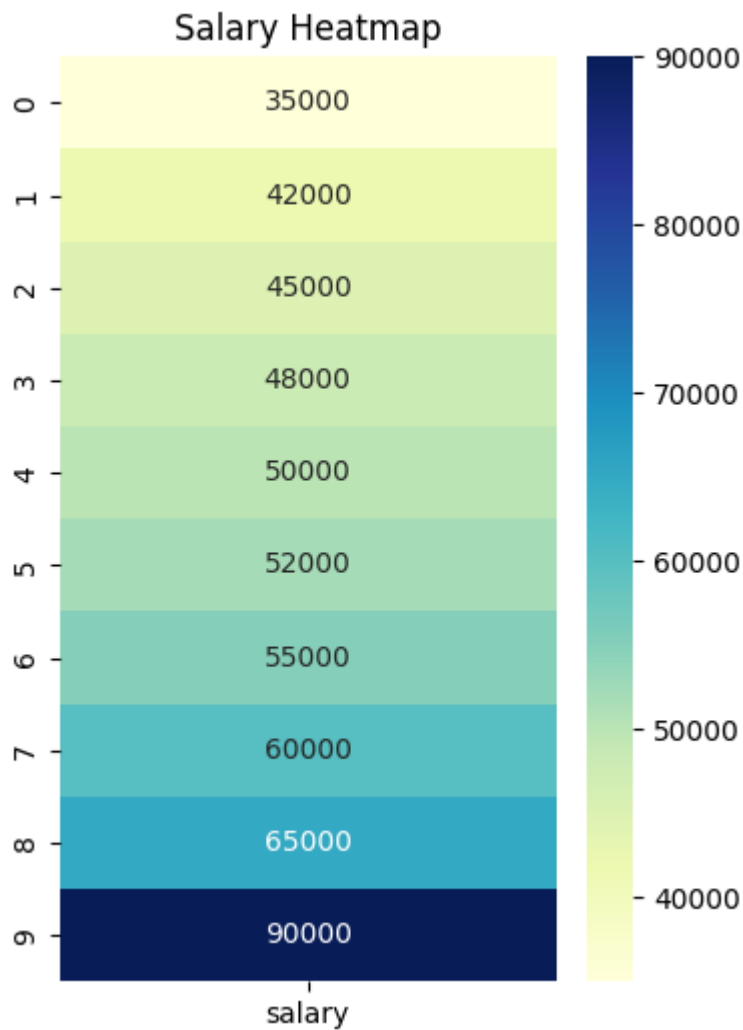
```

salary
0    35000
1    42000
2    45000
3    48000
4    50000
5    52000
6    55000
7    60000
8    65000
9    90000

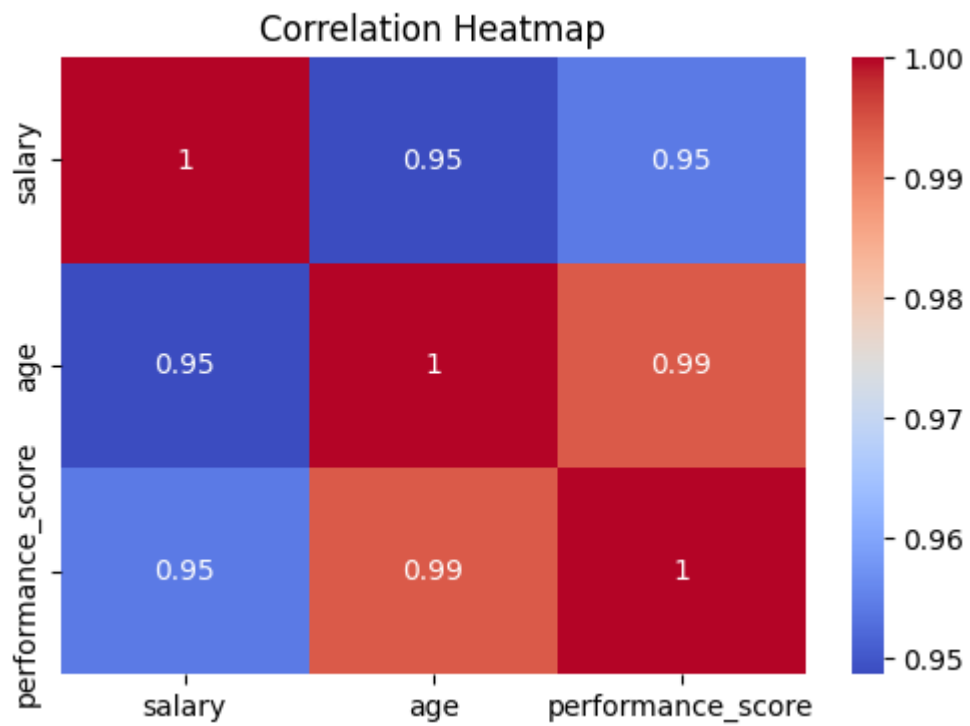
```



```
In [61]: plt.figure(figsize=(4, 6))
sns.heatmap(df, annot=True, cmap='YlGnBu', fmt='d')
plt.title('Salary Heatmap')
plt.show()
```

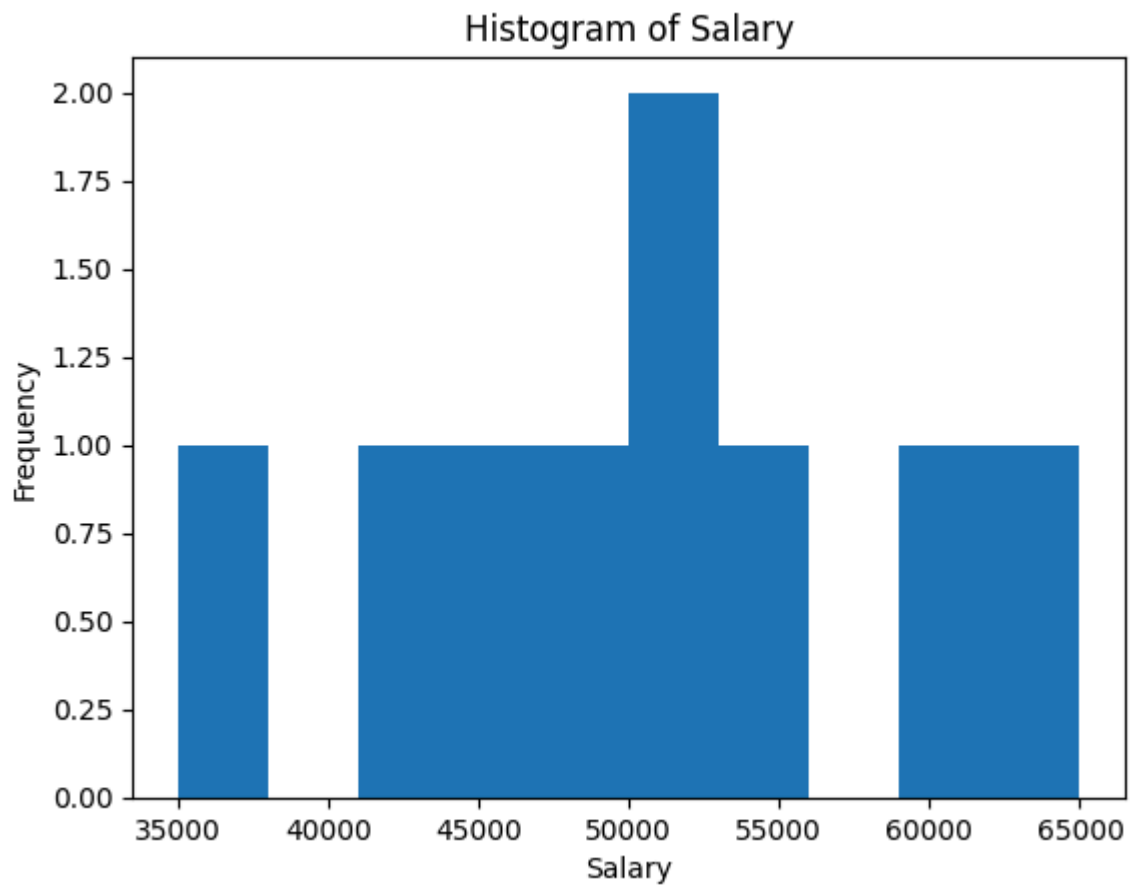


```
In [62]: df = pd.DataFrame({
    'salary': [35000, 42000, 45000, 48000, 50000, 52000, 55000, 60000, 65000, 90000],
    'age': [22, 25, 28, 30, 32, 35, 38, 40, 45, 50],
    'performance_score': [65, 70, 72, 75, 78, 80, 82, 85, 88, 95]})
plt.figure(figsize=(6, 4))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```



```
In [63]: df = pd.DataFrame({
    'salary': [35000, 42000, 45000, 48000, 50000,
               52000, 55000, 60000, 65000]
})

plt.hist(df['salary'])
plt.title('Histogram of Salary')
plt.xlabel('Salary')
plt.ylabel('Frequency')
plt.show()
```

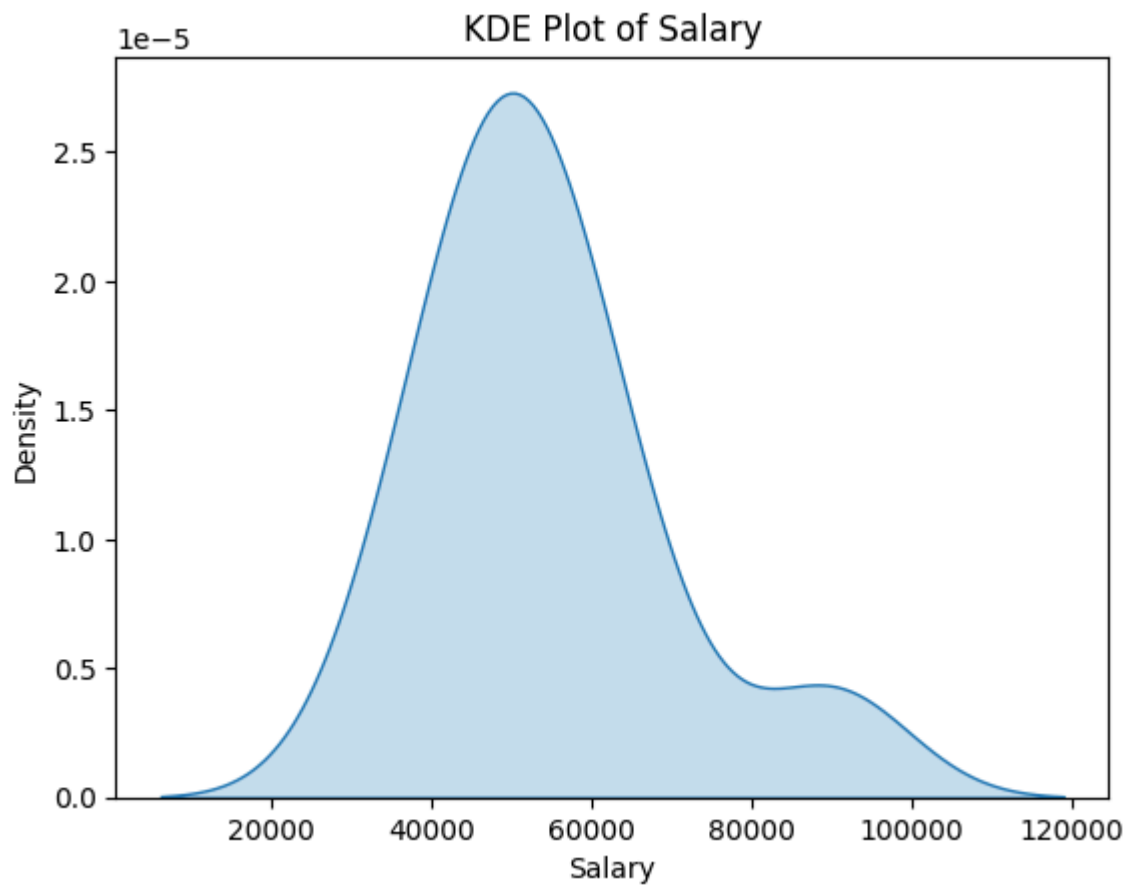


```
In [64]: df = pd.DataFrame({
    'salary': [35000, 42000, 45000, 48000, 50000,
               52000, 55000, 60000, 65000, 90000]
})

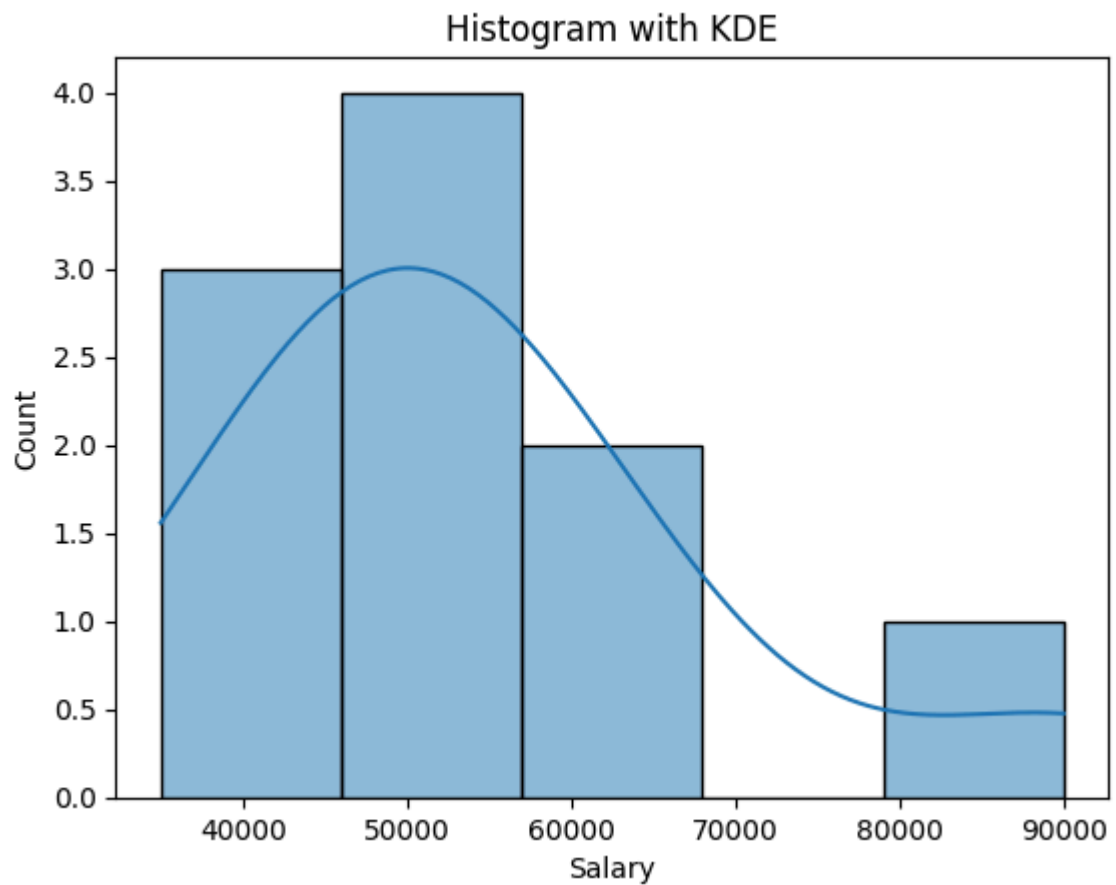
plt.hist(df['salary'], bins=5)
plt.title('Histogram of Salary (5 Bins)')
plt.xlabel('Salary')
plt.ylabel('Frequency')
plt.show()
```



```
In [66]: sns.kdeplot(data=df, x='salary', fill=True)
plt.title('KDE Plot of Salary')
plt.xlabel('Salary')
plt.ylabel('Density')
plt.show()
```



```
In [70]: sns.histplot(data=df, x='salary', kde=True)
plt.title('Histogram with KDE')
plt.xlabel('Salary')
plt.ylabel('Count')
plt.show()
```



```
In [71]: plt.plot([1,2,3],[10,20,30])
plt.title("Sales Trend")
plt.savefig("Sales Chart.png")
plt.show()
```

